

# Adversarial Vulnerability of Deep Learning - Based LPI Detectors

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## ABSTRACT

Deep learning detectors operating on time-frequency images (TFIs) achieve strong performance in detecting and recognizing low-probability-of-intercept (LPI) radar waveforms, yet their robustness to adversarial perturbations has received limited attention. This paper provides a modulation-wise evaluation of two canonical attacks—Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD)—against a convolutional neural networks based LPI detector under a black-box threat model, where transferable perturbations are crafted via surrogate models. Experiments cover twelve representative LPI modulations. Using per-modulation adversarial accuracy, attack success rate, and a TFI-domain perturbation budget  $\epsilon$ , we demonstrate both the effectiveness of adversarial examples and quantify the distortion-evasion trade-off relative to the original signals. The findings offer modulation-aware guidance on optimal attack settings, and highlight configurations that achieve maximal evasion with minimal perturbation.

**Key Words** : LPI radar, Adversarial examples, Time-frequency images

## I. Introduction

Low-Probability-of-Intercept (LPI) radars are designed to accomplish sensing objectives while minimizing the chance that an adversary can detect, classify, or geolocate the emission. This is typically achieved through waveform agility, low peak power with processing gain, and rapid parameter variation over time. As modern electronic support measures (ESM) have improved in sensitivity and coverage, the operational importance of LPI has increased, motivating continued innovations in waveform design and adaptive transmission policies<sup>[1]</sup>.

In parallel, systems tasked with detecting or identifying LPI signals have also advanced. Beyond traditional likelihood and constant false alarm rate (CFAR) based approaches, machine learning-based detectors that operate on time frequency images

(TFIs)—such as short time fourier transform (STFT) spectrograms—have shown strong accuracy and throughput by leveraging convolutional neural networks (CNNs). This TFI-centric paradigm is attractive because it converts diverse radar emissions into a common 2-D representation amenable to feature learning, enabling joint detection and modulation recognition across heterogeneous LPI waveforms<sup>[2-5]</sup>.

However, CNNs are known to be vulnerable to adversarial examples: carefully crafted, small perturbations to the input that induce erroneous model outputs. Such attacks are well documented in vision, audio, and natural language domains<sup>[6-8]</sup> and have recently been explored for radio signal classification as well<sup>[9]</sup>. Despite the clear security implications for spectrum sensing and electronic warfare, their applicability to LPI radar TFIs has received comparatively little systematic study. In particular, it

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remains unclear (i) how much the detection performance of TFI-based LPI detectors degrades under canonical adversarial methods, and (ii) how the effectiveness of adversarial attacks varies across different LPI modulation types under realistic radar transmission constraints.

This paper provides an empirical answer to those questions. We focus on FGSM (Fast Gradient Sign Method)<sup>[6]</sup> and PGD (Projected Gradient Descent)<sup>[7]</sup>, two widely used adversarial attack algorithms, as representative baselines. We evaluate their impact on a CNN-based LPI detector that ingests TFIs derived from a diverse set of LPI waveforms (twelve representative modulation schemes). Our analysis is conducted primarily under a black-box threat model: the attacker (radar transmitter) does not know the defender's model parameters but may train surrogate models to craft transferable perturbations. To reflect operational variability, we incorporate Expectation over Transformations (EOT) during adversarial generation, sampling STFT/normalization choices, SNR, and channel conditions so that perturbations remain effective under realistic preprocessing and propagation changes<sup>[8]</sup>.

Motivated by the need to understand the robustness of modern TFI-based detectors, this study pursues two goals: (i) to demonstrate the effectiveness of adversarial examples against LPI radar signal detection and (ii) to analyze the inherent trade-off between input distortion and detector evasion. Our primary metrics are per-modulation Adversarial Accuracy (AA), attack success rate (ASR), and the perturbation budget  $\epsilon$  relative to the no-attack baseline<sup>[10]</sup>. We investigate the effect of varying epsilon on AA and ASR across different attack methods and analyze performance variations by LPI radar modulation type. The main contributions of this work are threefold:

- Unlike prior RF adversarial studies that primarily focus on raw I/Q modulation classification or over-the-air symbol-level attacks, this work presents the first systematic evaluation of adversarial vulnerabilities in CNN-based LPI radar detectors operating directly on time-frequency images.

- Modulation-wise analysis. We report per-modulation AA, ASR, and the distortion-performance trade-off under FGSM/PGD<sup>[6-7]</sup>.
- Practitioner guidance. We provide actionable recommendations on modulation-aware attack settings, highlight limits of FGSM versus PGD in the low-distortion regime, and characterize modulation-specific vulnerabilities.

To situate our findings, we specify the scope and threat model as follows. Our analysis focuses on the TFI domain directly used by modern CNN-based detectors. The primary objective is untargeted evasion, aiming to reduce detection accuracy or induce misclassification among LPI classes. We consider a transfer-based black-box scenario, in which perturbations are crafted on a surrogate model trained with the same label space and subsequently evaluated on the target detector.

The remainder of the paper is organized as follows. Section 2 describes the LPI signal set, the TFI preprocessing pipeline, and the CNN detector. Section 3 formalizes the threat model and details FGSM/PGD alongside surrogate training. Section 4 presents the experimental setup, evaluation metrics, and results, analyzing per-modulation performance degradation and the distortion evasion trade-off. Section 5 concludes with findings and modulation-aware recommendations.

## II. Deep Learning-Based LPI Detection Model

### 2.1 LPI Signals

LPI radars employ specialized waveforms to evade detection and analysis by intercept receivers. Contemporary military systems increasingly diversify carrier frequency, modulation, and amplitude while maintaining low transmit power to further raise interception difficulty. Because LPI emissions typically exhibit low peak pulse power, wide occupied bandwidth, narrow beam patterns, and inconspicuous intra-pulse modulation, conventional Fast Fourier Transform (FFT)-based detection and classification approaches tend to be less effective.

We model an intercepted LPI radar emission in complex baseband as a narrowband carrier with a time-varying phase law corrupted by additive noise<sup>[1]</sup>,

$$x(t) = Ae^{j(2\pi f_c t + \phi(t))} + n(t), \quad (1)$$

where  $A$  is the amplitude,  $f_c$  the carrier frequency,  $\phi(t)$  the intra-pulse phase trajectory, and  $n(t)$  additive white Gaussian noise. This canonical form captures the signal part handled by the intercept receiver and the inevitable noise contribution. Beyond this parametric description, LPI waveforms are distinguished by properties tied to the ambiguity function and autocorrelation, which in turn govern range/velocity resolution and related sensing performance. Those characteristics depend on the chosen intra-pulse modulation, motivating a taxonomy by the quantity that is modulated within the pulse. In practice, we group LPI signals into frequency-modulated (FM) and phase-modulated (PM) families. FM waveforms modulate the instantaneous frequency term, as in linear FM (chirp), Costas frequency hopping, and nonlinear FM, whereas PM waveforms shape the phase trajectory while the carrier remains fixed, for example biphasic and polyphase codes. Equivalently, FM varies the frequency component under a time-invariant phase law, and PM varies the phase term under a constant carrier.

Following prior work in LPI waveform recognition, our study spans twelve representative intra-pulse modulations: two FM types (LFM and Costas) and ten PM types (BPSK, Frank, P1, P2, P3, P4, T1, T2, T3, T4)<sup>[11]</sup>. These classes cover widely used frequency-hopping, biphasic, and polyphase designs in LPI radar systems and are commonly employed in classification benchmarks. For comprehensive information on the modulation techniques and characteristics of these waveforms, please refer to<sup>[1]</sup>.

## 2.2 Time-Frequency Image

To capture discriminative structures in LPI radar emissions, each intercepted complex baseband signal is first mapped into a time-frequency (TF) representation and then rendered as a time-frequency

image. We employ the STFT, which offers the advantage of low computational complexity<sup>[12]</sup>. The STFT of a signal  $x(t)$  is defined as follows<sup>[13]</sup>:

$$STFT(\omega, \tau) = \int_{-\infty}^{\infty} e^{-j\omega t} w(t - \tau) x(t) dt. \quad (2)$$

Here,  $w(t - \tau)$  denotes the window function that specifies the range of frequency transformation. Accordingly,  $STFT(\omega, \tau)$  represents the Fourier transform of a localized segment of the signal  $x(t)$ , extracted using the window function  $w(t)$  centered at  $\tau$ . By discretizing the continuous STFT, the following expression is obtained:

$$STFT[\omega, l] = \sum_{n=-\infty}^{\infty} e^{-j\omega l} w[n-l] x[n]. \quad (3)$$

In this case,  $w[n]$  denotes the discretized form of  $w(t)$ , and the range is given by  $0 \leq n \leq N-1$ .

This STFT-based approach yields a unified two-dimensional representation across heterogeneous LPI modulations, exposing distinctive intra-pulse signatures such as chirp ridges, frequency hops, and phase-coded micro-patterns.

Such representations can be effectively exploited by deep learning models, including Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), to detect and classify modulation types<sup>[14]</sup>.

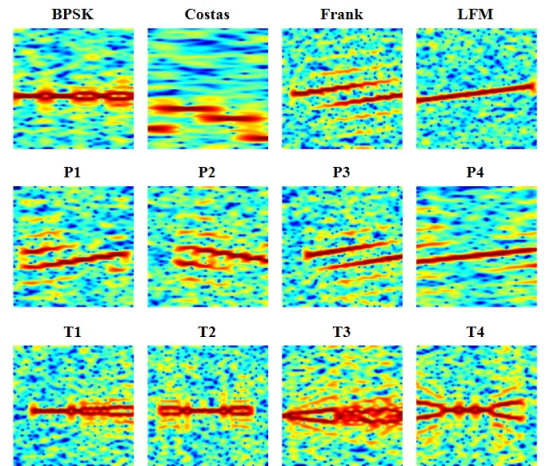


Fig. 1. STFT Time-Frequency Images for Twelve LPI Radar Modulations.

Representative time-frequency images (TFIs) for the twelve modulation classes considered in this work are illustrated in Figure 1, where each class exhibits a characteristic time-frequency texture that enables reliable recognition.

### 2.3 LPI Detection Model

Our detector consumes single-channel TFIs derived from complex I/Q records and predicts one of 12 intra-pulse modulation classes. The raw I/Q records are transformed into log-magnitude spectrograms via an STFT pipeline, followed by dynamic-range clipping and per-image normalization to  $[0, 1]$ . The resulting grayscale TFIs are resized to  $224 \times 224$  and fed directly to the network at inference.

To reduce overfitting to any particular preprocessing choice, we mildly vary the STFT configuration during training, including sampling window length, hop size, FFT size, and the clipping/normalization range, while holding the configuration fixed for validation and all adversarial evaluations to ensure fair comparisons.

The baseline backbone is ResNet-50 with a 1-channel input stem, global average pooling, and a 12-way softmax classifier. We train with multiclass cross-entropy and light label smoothing ( $\epsilon_s = 0.1$ ), optimized by AdamW (initial learning rate  $3 \times 10^{-4}$ , weight decay 0.05) under a cosine schedule with a 5-epoch warm-up, for a total of 80 epochs. The batch size is 64, and we apply small dropout in the classifier head. Augmentations are semantics-preserving (e.g., mild TF crop/pad and weak Gaussian noise). The corpus is split 80/10/10 into train/validation/test per class and SNR level, and early stopping on validation accuracy/F1 is used to limit overfitting. All experiments use PyTorch/torchvision; TFIs are passed to the model as 1-channel images subsequently normalized to  $[-1, 1]$  by an input transform.

## III. Adversarial Attacks

We formalize and evaluate adversarial attacks against the TFI-based LPI detector introduced in Section II. In keeping with operational constraints, we adopt the black-box threat model as our default: the

attacker does not know the detector's architecture, parameters, or training loss, and can only observe outputs (labels or scores). Throughout, the attack objective is untargeted evasion, aiming to reduce detection accuracy or induce any misclassification, rather than forcing a specific wrong class<sup>[15]</sup>.

Inputs are grayscale TFIs of size  $224 \times 224$  scaled to the pixel range  $[0, 1]$ . All perturbations are applied in the pixel domain before normalization: given an image  $x$  and perturbation  $\delta$ , we form  $\tilde{x} = \text{clip}_{[0,1]}(x + \delta)$  and then apply the detector's (fixed) normalization. We consider norm-bounded perturbations and use  $L_\infty$  as the default constraint  $\|\delta\|_\infty \leq \epsilon$ , with budget  $\epsilon$ . For completeness, we also implemented  $L_2$ -bounded variants; unless otherwise noted, all reported results refer to  $L_\infty$ . For fair comparisons across methods, we sweep  $\epsilon$  over a preset grid.  $L_\infty$  constrains the maximum per-pixel deviation, while  $L_2$  constrains the total perturbation energy; because these norms capture different notions of distortion, we analyze them separately.

We instantiate two canonical first-order methods. FGSM<sup>[6]</sup> performs a single step in the direction of the loss gradient's sign:

$$x_{adv} = \text{clip}(x + \epsilon \text{sign}(\nabla_x L(f(x), y))), \quad (4)$$

where  $f$  is the model and  $\text{clip}$  denotes clipping to  $[0, 1]$ . FGSM is computationally inexpensive (one backward pass) but typically less robust to preprocessing changes. PGD<sup>[7]</sup> iteratively maximizes the loss under the constraint. Starting from  $x_0$  (optionally a random start inside the  $\epsilon$ -ball), it repeats

$$x^{(k+1)} = \Pi_{B_\infty(x, \epsilon)}(x^{(k)} + \alpha \text{sign}(\nabla_{x^{(k)}} L(f(x^{(k)}), y))), \quad (5)$$

for  $k = 0, \dots, K-1$ , with step size  $\alpha$ , iteration count  $K$ , and projection  $\Pi_{B_\infty(x, \epsilon)}$  onto the chosen  $L_p$  ball. For  $L_2$ , the sign step is replaced by a normalized gradient step. PGD generally achieves higher attack success at a given  $\epsilon$  but costs  $K$  backward passes. For both FGSM and PGD we use the same  $\epsilon$  sweep

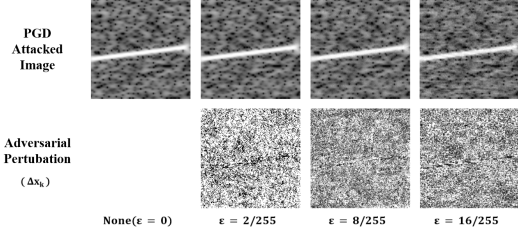


Fig. 2. Visualization of  $L_\infty$ -bounded PGD attacks at  $\epsilon \in \{0, 2, 8, 16\}/255$ . Top: attacked TFIs. Bottom: contrast stretched perturbations  $\delta$ .

(e.g.,  $\{2, 4, 6, 8, 10, 12, 14, 16\}/255$  in  $L_\infty$ ) to trace the distortion-evasion trade-off. FGSM applies a single projection/clipping; PGD projects and clips at every iteration. Figure 2 illustrates PGD examples at  $\epsilon \in \{0, 2, 8, 16\}/255$ .

Under black-box constraints we craft perturbations using a surrogate (attacker) model trained to mimic the defender’s TFI pipeline and label space, then transfer the examples to the (unknown) detector. To improve transferability and reduce brittleness to receiver-side variability, we employ EOT<sup>[16]</sup>:  $\Gamma$  denotes a predefined distribution over weak, semantics-preserving transforms intended to capture receiver-side variability (e.g., small time-frequency shifts, light crop/pad, mild dynamic-range rescaling, weak additive noise, and small STFT parameter jitters). At each attack step we draw  $T \sim \Gamma$  and optimize the expected loss under these sampled transforms, which improves transferability and robustness to preprocessing mismatches.

$$\max_{\|\delta\| \leq \epsilon} E_{T \sim \Gamma} [L(F(T(x+\delta)), y)]. \quad (6)$$

Gradients are taken through the sampled transforms when differentiable. At evaluation time the defender’s preprocessing is held fixed.

We report the following metrics for each combination of attack method, norm, and  $\epsilon$ . Specifically, we evaluate performance using three key metrics. Clean Accuracy (CA) refers to the accuracy on the original, unperturbed inputs. AA denotes the accuracy after applying the perturbation. ASR measures the proportion of inputs that were correctly classified before the attack but misclassified afterward

—that is, the error rate conditioned on the clean-correct subset.

We additionally provide per-modulation breakdowns to reveal which waveform types are most vulnerable. All experiments use identical  $\epsilon$  sweeps for FGSM and PGD and apply the perturbation in the TFI pixel domain with clipping to  $[0, 1]$  prior to normalization.

#### IV. Experiments and Analysis

This section presents the empirical results evaluating the vulnerability of the TFI-based LPI radar detector to adversarial perturbations generated using FGSM and PGD. Under clean conditions, the test set is class-balanced across all twelve modulation types. Accordingly, the detector achieves a test accuracy of 94.8%, with a comparable macro-averaged F1 score of 93.7%, confirming consistent performance across modulation classes. All experiments are conducted under a black-box threat model, where transferable adversarial examples are crafted via a surrogate model with EOT-based transformation robustness. To quantify the impact of adversarial attacks, we vary the perturbation budget  $\epsilon$  and report two key performance metrics: AA and ASR. These metrics provide complementary views of the detector’s degradation, capturing both absolute classification accuracy and the fraction of successfully corrupted predictions.

Figure 3 illustrates the effect of increasing  $\epsilon$  on AA and ASR for both FGSM and PGD. As  $\epsilon$  increases, AA consistently decreases while ASR increases, reflecting the growing effectiveness of the perturbations. The performance drop is more severe for PGD, highlighting its superior optimization process compared to the single-step FGSM. For instance, with  $\epsilon = 10/255$ , PGD already causes AA to drop below 70%, whereas FGSM shows a more gradual decline. This distortion-evasion trade-off confirms that even small  $L_\infty$ -bounded perturbations can significantly undermine the detector’s reliability. The trend is further supported by the ASR curves: PGD shows a noticeably steeper increase in ASR than

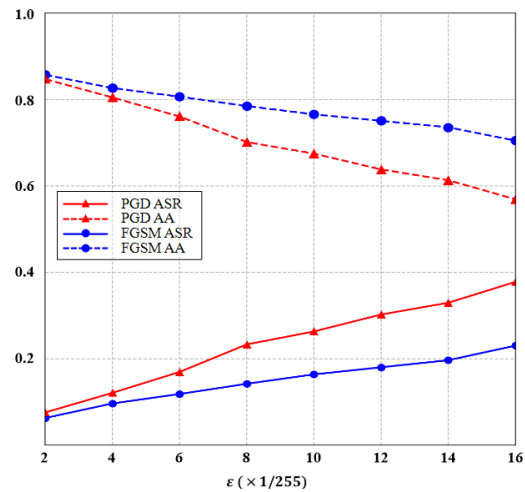


Fig. 3. AA and ASR as a function of perturbation budget  $\epsilon$  for FGSM and PGD attacks.

FGSM as  $\epsilon$  grows. At  $\epsilon = 16/255$ , PGD achieves over 35% ASR, whereas FGSM remains below 25%.

These results indicate that PGD is more effective at inducing misclassifications under the same perturbation constraints and poses a more serious threat to the detector’s robustness.

Figure 4 provides a modulation-wise comparison of ASR under FGSM and PGD attacks at  $\epsilon = 8/255$ . The figure shows that all twelve LPI waveform classes are affected, but to varying degrees. Certain modulations, such as P3, Frank, and BPSK, exhibit notably higher ASR under both attacks, suggesting these modulation types are particularly vulnerable to

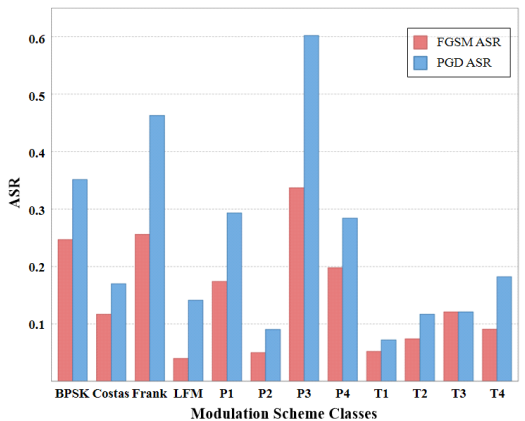


Fig. 4. Modulation-wise ASR comparison between FGSM and PGD attacks( $\epsilon = 8/255$ ).

small input perturbations.

In contrast, LFM and P2 waveforms show relatively lower ASR, likely due to their more stable and less distinctive time-frequency structures. These observations suggest that intra-pulse structure and spectral density directly influence the adversarial robustness of different waveforms. Notably, PGD yields consistently higher ASR than FGSM for every modulation, reaffirming its greater attack strength.

To further analyze how adversarial perturbations affect classification behavior, we present a confusion matrix in Table 1, generated under a PGD attack with  $\epsilon = 8/255$ . Each cell in the matrix reports the classification accuracy from a true class (column) to a predicted class (row), with the clean (pre-attack) accuracy shown outside parentheses and the adversarial (post-attack) accuracy inside. The diagonal entries represent correct classifications. Across nearly all classes, these values decrease significantly under attack, reflecting the general degradation of the detector’s reliability. The off-diagonal entries, conversely, become more populated, indicating increased confusion between modulation types. For example, P3 waveforms are frequently misclassified as Frank, suggesting structural similarity in their TFI representations. P1 waveforms, meanwhile, are not significantly misclassified as Frank or P2, but instead show consistent confusion with P4, likely due to overlapping phase-coded patterns. This indicates that PGD attacks do not cause purely random misclassifications, but instead induce systematic errors that align with underlying waveform similarities. Overall, Table 1 provides a fine-grained view of class-specific vulnerabilities, both in isolation and in relation to other classes. Such insights are crucial for understanding where defense mechanisms—such as adversarial training or class-specific regularization—should be applied to enhance model robustness in adversarial environments.

These experimental results validate the central hypothesis of this study: that TFI-based CNN detectors for LPI radar waveforms are highly susceptible to adversarial attacks, even in a black-box setting. Moreover, the effectiveness of the attack and the degree of vulnerability vary significantly by

Table 1. Confusion matrix before and after PGD attack at  $\epsilon = 8/255$ .

|        | BPSK           | Costas         | Frank          | LFM            | P1             | P2             | P3             | P4             | T1             | T2             | T3             | T4             |
|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| BPSK   | 95.1<br>(31.4) | 2.9<br>(2.7)   | 0.0<br>(0.8)   | 0.0<br>(0.0)   | 0.0<br>(2.2)   | 0.0<br>(2.6)   | 0.0<br>(0.0)   | 0.7<br>(2.8)   | 1.3<br>(37.3)  | 0.0<br>(18.1)  | 0.0<br>(1.2)   | 0.0<br>(0.9)   |
| Costas | 2.2<br>(8.9)   | 94.4<br>(68.7) | 0.9<br>(5.7)   | 0.0<br>(0.1)   | 0.7<br>(8.3)   | 0.9<br>(2.7)   | 0.0<br>(0.0)   | 0.0<br>(0.3)   | 0.0<br>(2.2)   | 0.0<br>(0.0)   | 0.0<br>(1.8)   | 0.9<br>(1.3)   |
| Frank  | 1.3<br>(1.2)   | 0.0<br>(0.3)   | 79.4<br>(35.6) | 0.0<br>(1.9)   | 0.7<br>(0.7)   | 1.4<br>(2.6)   | 14.5<br>(53.6) | 0.0<br>(1.3)   | 0.5<br>(0.4)   | 0.0<br>(0.0)   | 2.2<br>(2.4)   | 0.0<br>(0.0)   |
| LFM    | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 99.4<br>(81.6) | 0.0<br>(0.9)   | 0.0<br>(0.0)   | 0.0<br>(0.3)   | 0.6<br>(14.2)  | 0.0<br>(0.0)   | 0.0<br>(0.8)   | 0.0<br>(2.2)   | 0.0<br>(0.0)   |
| P1     | 0.0<br>(0.0)   | 0.1<br>(0.6)   | 1.2<br>(0.1)   | 0.0<br>(0.8)   | 87.8<br>(45.7) | 0.9<br>(1.0)   | 0.0<br>(7.3)   | 6.7<br>(37.6)  | 0.0<br>(0.0)   | 0.2<br>(0.0)   | 2.2<br>(0.5)   | 4.5<br>(0.0)   |
| P2     | 0.0<br>(2.1)   | 0.0<br>(0.6)   | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 99.5<br>(85.9) | 0.0<br>(1.1)   | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 0.0<br>(1.5)   | 0.5<br>(2.3)   | 0.0<br>(6.5)   |
| P3     | 1.2<br>(0.4)   | 0.0<br>(1.4)   | 7.2<br>(57.1)  | 0.0<br>(0.0)   | 2.0<br>(3.8)   | 0.9<br>(0.6)   | 87.3<br>(33.3) | 0.0<br>(0.9)   | 0.0<br>(0.6)   | 0.0<br>(0.0)   | 0.9<br>(1.3)   | 0.5<br>(0.6)   |
| P4     | 1.0<br>(0.9)   | 0.8<br>(1.3)   | 0.0<br>(2.8)   | 1.9<br>(12.2)  | 6.2<br>(23.9)  | 0.0<br>(2.1)   | 0.0<br>(0.0)   | 87.3<br>(54.1) | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 2.3<br>(2.1)   | 0.5<br>(0.6)   |
| T1     | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 0.0<br>(0.9)   | 0.0<br>(1.7)   | 0.0<br>(1.3)   | 0.0<br>(0.0)   | 0.0<br>(0.1)   | 0.0<br>(0.2)   | 97.3<br>(86.9) | 0.5<br>(4.8)   | 0.8<br>(2.0)   | 1.4<br>(2.1)   |
| T2     | 0.0<br>(2.7)   | 0.0<br>(0.0)   | 0.0<br>(0.2)   | 0.5<br>(3.6)   | 0.0<br>(0.0)   | 0.0<br>(0.0)   | 0.0<br>(0.1)   | 0.0<br>(0.0)   | 1.1<br>(4.3)   | 90.9<br>(74.4) | 2.1<br>(3.7)   | 5.4<br>(11.0)  |
| T3     | 0.0<br>(0.0)   | 0.2<br>(0.3)   | 0.0<br>(1.9)   | 0.0<br>(1.3)   | 0.8<br>(0.0)   | 0.0<br>(0.8)   | 0.0<br>(1.0)   | 0.0<br>(2.3)   | 2.3<br>(8.7)   | 0.6<br>(3.4)   | 93.2<br>(74.1) | 2.9<br>(6.2)   |
| T4     | 0.0<br>(0.0)   | 0.0<br>(0.2)   | 1.3<br>(0.8)   | 0.0<br>(1.1)   | 1.9<br>(1.8)   | 0.0<br>(4.8)   | 0.0<br>(0.0)   | 2.1<br>(3.3)   | 1.0<br>(4.3)   | 1.8<br>(4.6)   | 2.4<br>(2.8)   | 89.5<br>(76.3) |

modulation type. PGD consistently outperforms FGSM in terms of both ASR and degradation of AA, especially at moderate  $\epsilon$  levels. These findings highlight the need for modulation-aware robustness evaluations and more resilient TFI processing pipelines in deep-learning-based electronic warfare systems.

V. Conclusion

This paper investigated the adversarial vulnerability of deep learning-based LPI radar detectors operating on time - frequency images, extending prior RF adversarial studies by focusing on LPI-specific waveforms and TFI-based CNN detectors under a transfer-based black-box threat model. While CNN-based models perform well under clean conditions, our experiments show that even small perturbations crafted using FGSM and PGD can significantly degrade detection accuracy, particularly

for certain modulation types. PGD consistently achieved higher ASRs than FGSM at the same perturbation levels, indicating greater attack effectiveness with equivalent distortion. Modulation-wise analysis further revealed that waveforms such as P3, Frank, and BPSK are especially susceptible to adversarial attacks, while LFM and P2 showed greater robustness. The confusion matrix results also demonstrated increased misclassifications and systematic confusion among structurally similar waveforms. These findings highlight the importance of evaluating and improving the robustness of TFI-based LPI detectors. Future research may focus on developing defense mechanisms such as adversarial training and modulation-aware regularization to enhance the resilience of radar signal classifiers in adversarial environments.

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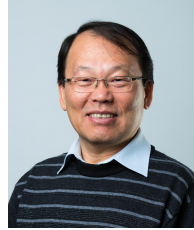
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